

Encoding Ambiguity in Logical Form

Comprehensive Notes in Formal Semantics

Introduction

Ambiguity is a pervasive and systematic feature of natural language. A single sentence may express more than one meaning, even when its pronunciation and surface syntax remain constant. Formal semantics seeks to explain ambiguity not as a defect of language but as a structural consequence of the compositional architecture of grammar. The key theoretical claim is that ambiguity arises when a single surface string corresponds to multiple logical forms (LFs). Logical Form is an abstract syntactic representation at which semantic interpretation is computed. By encoding ambiguity at LF, we preserve the principle of compositionality while explaining how multiple interpretations are generated.

Ambiguity may arise from different structural attachments, different quantifier scope configurations, lexical polysemy, negation interaction, or intensional embedding. In each case, formal semantics encodes the ambiguity by assigning distinct hierarchical representations. The following sections provide a systematic exploration of how ambiguity is encoded in logical form, with detailed step-by-step derivations.

1. Structural Ambiguity

Structural ambiguity arises when a sentence has more than one syntactic parse, each corresponding to a different semantic interpretation.

Consider:

Old men and women

Two structural parses are possible:

1. Old modifies both men and women.

2. Old modifies only men.

Lexical entries:

$$\begin{aligned}old &= \lambda x Old(x) \\men &= \lambda x Man(x) \\women &= \lambda x Woman(x)\end{aligned}$$

Interpretation 1:

$$\lambda x (Old(x) \wedge (Man(x) \vee Woman(x)))$$

Interpretation 2:

$$\lambda x ((Old(x) \wedge Man(x)) \vee Woman(x))$$

The ambiguity arises from different structural grouping, and logical form explicitly encodes these alternative structures.

2. Quantifier Scope Ambiguity

Quantifier scope ambiguity occurs when multiple quantifiers can take different hierarchical positions at LF.

Consider:

Every student read a book.

Reading 1: Universal Wide Scope

$$\forall x (Student(x) \rightarrow \exists y (Book(y) \wedge Read(x, y)))$$

Step-by-step compositional derivation:

$$student = \lambda x Student(x)$$

$$\begin{aligned}
book &= \lambda y Book(y) \\
read &= \lambda y \lambda x Read(x, y) \\
every &= \lambda P \lambda Q \forall x (P(x) \rightarrow Q(x)) \\
a &= \lambda P \lambda Q \exists y (P(y) \wedge Q(y))
\end{aligned}$$

Composition yields:

$$\forall x (Student(x) \rightarrow \exists y (Book(y) \wedge Read(x, y)))$$

Reading 2: Existential Wide Scope

$$\exists y (Book(y) \wedge \forall x (Student(x) \rightarrow Read(x, y)))$$

The difference arises from alternative scope positions at LF.

3. Negation Scope Ambiguity

Consider:

Every student didn't pass.

Two interpretations:

Reading 1: Negation Wide Scope

$$\neg \forall x (Student(x) \rightarrow Passed(x))$$

Equivalent to:

$$\exists x (Student(x) \wedge \neg Passed(x))$$

Reading 2: Universal Wide Scope

$$\forall x (Student(x) \rightarrow \neg Passed(x))$$

Logical form encodes which operator has hierarchical dominance.

4. Attachment Ambiguity

Consider:

John saw the man with the telescope.

Instrumental Reading:

$$\exists x(Man(x) \wedge Saw(j, x) \wedge With(j, telescope))$$

Modifier Reading:

$$\exists x(Man(x) \wedge With(x, telescope) \wedge Saw(j, x))$$

The ambiguity arises from alternative syntactic attachment sites.

5. Lexical Ambiguity

Consider:

John went to the bank.

Two lexical denotations:

$$WentTo(j, Bank_{financial})$$

$$WentTo(j, Bank_{river})$$

Lexical ambiguity corresponds to distinct predicate interpretations.

6. De Dicto and De Re Ambiguity

Intensional contexts introduce scope ambiguity between quantifiers and modal or attitude operators.

Consider:

John wants to marry a doctor.

De Dicto:

$$Wants(j, \exists x(Doctor(x) \wedge Marry(j, x)))$$

De Re:

$$\exists x(Doctor(x) \wedge Wants(j, Marry(j, x)))$$

The difference arises from the relative position of the existential quantifier and the intensional operator.

7. Coordination Ambiguity

Consider:

Every student and every teacher left.

Logical encoding:

$$(\forall x(Student(x) \rightarrow Left(x))) \wedge (\forall y(Teacher(y) \rightarrow Left(y)))$$

Different distributive or collective interpretations may arise depending on grouping.

8. Lambda-Based Derivation of Ambiguity

Lambda abstraction makes scope interactions explicit.

Example:

$$(\lambda P \lambda Q \forall x(P(x) \rightarrow Q(x)))(\lambda x Student(x))$$

Ambiguity arises when alternative applications reorder quantifiers before reduction.

Beta reduction step-by-step reveals how different hierarchies yield different meanings.

9. General Principles

Ambiguity arises when:

- Multiple hierarchical structures are possible.
- Quantifiers compete for scope.
- Operators interact differently.
- Lexical denotations vary.
- Intensional contexts alter substitution conditions.

Each ambiguity corresponds to a distinct logical form.

Conclusion

Encoding ambiguity in logical form reveals the deep structural basis of multiple interpretations in natural language. Rather than treating ambiguity as accidental, formal semantics shows that it arises from systematic interactions between syntactic hierarchy and semantic composition. Logical form provides the level of representation at which these differences become explicit. Through compositional derivation, type theory, and lambda abstraction, ambiguity is not merely observed but formally explained. This demonstrates the explanatory power of formal semantic theory in modeling the complexity of natural language meaning.